

SAGAR SINGH BHOLA

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EDUCATION

University of Michigan, Ann Arbor

December 2025

Bachelor of Science in Computer Science

GPA: 3.75

- **Relevant Coursework:** Data Structures and Algorithms, Computer Networks, Computer Security, Web Systems, Computer Architecture, User Interface Development, Human-Centered Software

TECHNICAL SKILLS

Languages/Technologies: Python, C++, C, JavaScript, TypeScript, SQL, Java, HTML, CSS

Frameworks/Libraries: React, Next.js, Vue, Node.js, Express.js, Flask, FastAPI, Pandas

Tools/Platforms: MySQL, Docker, AWS, Git, Postman, Playwright, Figma

EXPERIENCE

Data Engineering Co-op

May 2025 – Present

Ciena Healthcare

Southfield, MI

- **Engineered reusable ETL pipelines** using Python, Pandas, and SQL to automate preparation of 50+ Michigan Medicaid cost reports, reducing manual workload by **90%** and avoiding **\$100,000+** in external costs.
- **Developed a Playwright automation suite** to validate, synchronize, and securely upload positive pay files to the CIBC banking portal, reducing manual handling and upload errors.
- **Reduced pipeline delivery time by 60%** by working with the Senior Financial Controller to refine requirements and implement validation checks and audit logging across reporting workflows.

Lead Backend Engineer

Mar 2025 – Present

YesNoFingers (University of Michigan Language Resource Center)

Ann Arbor, MI

- **Architected a cloud-based speech and translation pipeline** using OpenAI Whisper, AWS Translate, and EC2, incorporating historical, team-curated datasets to process culturally specific medical terminology across 40+ languages.
- **Strengthened backend security and reliability** through input validation, API security testing, and end-to-end testing of translation workflows.
- **Conducted pilot testing with clinicians** at a Detroit-area free clinic, using feedback to improve backend request handling and privacy-conscious processing of sensitive medical content.

PROJECTS

AI Invoice Processor | *Next.js, TypeScript, FastAPI, Python, SQLite, Local LLM*

- **Built a local-LLM invoice processing pipeline** to extract invoice and line-item data from uploaded PDFs while keeping **100% of document processing local**.
- **Designed a FastAPI and SQLite data layer** to manage invoice records, attachments, extraction updates, review states, dashboard metrics, and accounting distributions.
- **Developed an AP review dashboard** with status-based queues, editable fields, draft persistence, embedded PDF previews, and CRUD configuration across **4 administrative categories**: vendors, facilities, GL accounts, and authorized users.

TCG Portfolio Evaluator | *React, TypeScript, Express.js, MySQL, Pokémon TCG API*

- **Developed a full-stack portfolio management platform** for card search, collection tracking, quantity management, and grid/table views using React, TypeScript, Express.js, and MySQL.
- **Integrated the Pokémon TCG API** to retrieve card metadata, images, and market pricing through **6+ RESTful endpoints** and duplicate-aware database upserts.
- **Built SQL-backed portfolio analytics** to calculate **3 collection metrics**: total market value, unique card holdings, and total card quantity.

Instagram Clone | *React, Python, Flask, SQL, AWS*

- **Developed a full-stack social media platform** using React, Flask, SQL, and AWS with a RESTful API for posts, comments, likes, and user sessions.
- **Engineered a paginated infinite-scroll feed** using JavaScript and the Fetch API to load content asynchronously and reduce the initial response payload.
- **Implemented 2 real-time social interactions**: double-click likes and instant commenting with immediate client-side updates.